

Appendix: unpaired PYTHIA $\rightarrow$ JEWEL jet translation, a first pilot

Question. Can the existing unpaired methods map a *clean* PYTHIA jet image to an output with held-out JEWEL statistics while keeping a meaningful dependence on the input jet? There is no matched JEWEL jet per PYTHIA jet: any unpaired map is one choice among many with the same marginals. Nothing here is a per-event medium modification.

Data (own caches, splits, normalisation)

- parents = (generator file, event): PYTHIA 2.64M, JEWEL 870k clean images; 7.2% of the PYTHIA images are exact copies across files, dropped
- one leading jet per parent, the closure's code for both: cone ≥ 10 GeV, axis $|\eta| < 0.73$, the 53 R = 0.4 towers in GeV on 16×16 ; no per-jet normalisation
- by parent: PYTHIA 200k / 10k / 20k, JEWEL 200k / 10k / 20k + 20k ref; the old JEWEL test jets reserved
- the crop drops half the event energy (out-of-cone radiation, recoil, the other jet)
- PYTHIA's cone E starts near 20 GeV, JEWEL's at 10: generators and selections differ, not only the medium

Models (seed 0, UVCAN-S backbone, 2 A6000 GPU h each, final checkpoint)

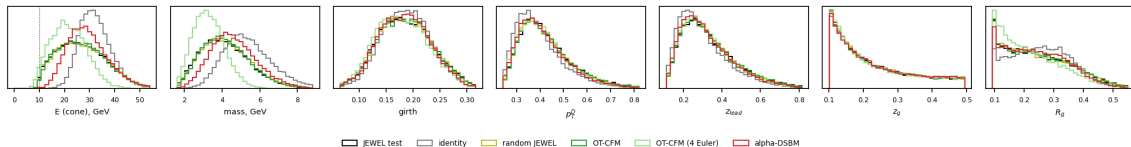
- **OT-CFM**: exact minibatch OT (256), squared L^2 of the standardised $\log(E + 0.1)$ canvases, straight path; ODE midpoint, NFE fixed on validation (128); 4 Euler*: speed only
- **α -DSBM**: independent-pair bridge 0.5 h, online refinement 1.5 h (rollouts in the budget), $\varepsilon = 0.25$, SDE 30 steps; one sample per jet (8 on 1000 jets)
- no new backbone, conditioning, shape or conservation loss, sweep, or test-data choice

Baselines: identity (the PYTHIA jet); random JEWEL (training jets, input ignored); JEWEL ref (held-out JEWEL: the finite-sample floor).

Read-out fixed before training: W1 to JEWEL test with bootstrap errors, input correlations, own-input preference, artefact flags.

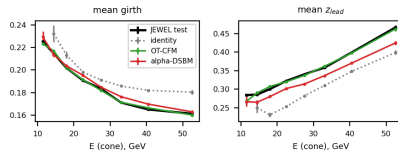
Distributions: **OT-CFM** at the JEWEL-vs-JEWEL floor, **α -DSBM** halfway

Fixed-crop observables, 20k test jets each; shapes after $E \geq 10$ GeV on every sample (E: all outputs)



	pass	E	mass	girth	p_T^D	z_{lead}	z_g	R_g
JEWEL ref	1.000	0.011	0.014	0.009	0.018	0.019	0.012	0.008
random JEWEL	1.000	0.019 ^o	0.024 ^o	0.011 ^o	0.010 ^o	0.011 ^o	0.005 ^o	0.010 ^o
identity	1.000	0.546	0.881	0.105	0.264	0.232	0.034 ^o	0.158
OT-CFM	0.995	0.025^o	0.021^o	0.012^o	0.010^o	0.008^o	0.017 ^o	0.012^o
OT-CFM , 4 Euler*	0.974	0.510	0.678	0.129	0.125	0.099	0.010 ^o	0.249
α-DSBM	0.999	0.256	0.325	0.056	0.112	0.099	0.036 ^o	0.073

W1/ σ to 20k held-out JEWEL test jets (bootstrap sd 0.004-0.011); every sample cut at cone $E \geq 10$ GeV, pass = share of outputs kept (JEWEL: 1 by construction). **Bold**: below the identity by > 3 sd; ^o: within 3 sd of the JEWEL-ref floor. *speed readout only.

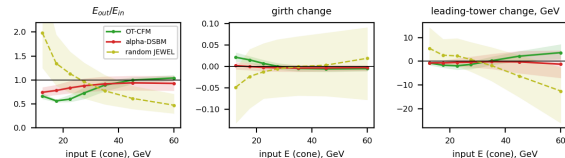


- **joint**, W1/ σ in E bins 10-20 / 20-30 / 30-60 GeV: girth **OT-CFM** 0.023 / 0.012 / 0.020, z_{lead} 0.030 / 0.025 / 0.016 (floor 0.02-0.04; identity 0.16-0.47; **α -DSBM** 0.07-0.18)
- **refund** anti- k_T jets ($p_T \geq 10$ GeV): **OT-CFM** within the floor in all 7 (pass 0.978, JEWEL 0.976); **α -DSBM** 0.03-0.34
- **α -DSBM** closes 51-64% of the identity's gap; 4 Euler steps barely move E

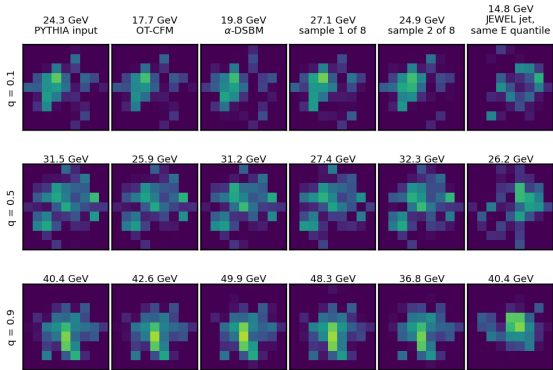
Each output keeps its input's layout; the energy change follows the spectra

	correlation, input vs output					own input preferred
	E	girth	mass	core	lead	
identity	1.00	1.00	1.00	1.00	1.00	1.000
random JEWEL	0.01	-0.00	0.01	-0.00	-0.00	0.498
OT-CFM	0.94	0.99	0.91	0.99	0.92	1.000
OT-CFM, 4 Euler*	0.86	0.99	0.82	0.99	0.89	1.000
α -DSBM	0.80	0.98	0.88	0.98	0.79	1.000

20k test jets, no selection (bootstrap sd of a correlation ≤ 0.003).
Own input preferred: the output is closer (normalised-shape EMD) to its input than to another test jet. These describe the maps, not errors.



- OT-CFM E_{out}/E_{in} (median): 0.60 at 20-25 GeV, 0.89 at 30-40, 1.00-1.04 above 40 GeV: the two samples' spectra (PYTHIA from 20 GeV, JEWEL from 10), not an energy-loss mechanism; hard jets get a harder core, as JEWEL's have.
 α -DSBM: 0.84-0.94, flatter



Fixed inputs (first 1000 test jets at cone- E quantiles q), one output each; α -DSBM: its output and 2 of 8 samples. Two samples of one jet differ more (shape EMD 0.068) than a sample differs from its input (0.056); per-jet energy spread $0.37 \sigma_E$: algorithmic variability, not a physical uncertainty.

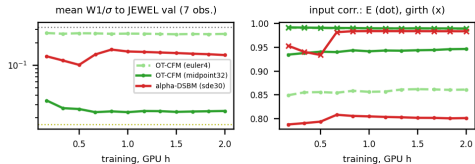
What the pilot establishes, what it does not, and cost

	GPU h	updates	upd./s	overhead	NFE	ms/jet
OT-CFM	2.00	122.7k	17.0	10% OT	128	6.58
4 Euler*					4	0.21
α-DSBM pretrain	0.50	26.1k	14.5	—		
+ refine	1.50	8.8k	1.62	89% roll-out	30	1.55

One A6000 each; GPU h: training time, rollouts included; peak memory 3.5-3.8 GB; latency at batch 2000. Solver on validation: **OT-CFM** 32 vs 64 NFE moves a jet's energy by 8% of its change (rule 1%), 64 vs 128 1.6%, 128 vs 256 0.5% $\Rightarrow$ 128. **α -DSBM** 30 vs 60 steps, same noise: 0.55 GeV vs 6.0 between samples $\Rightarrow$ 30.

Measured (seed 0, 20k held-out jets each)

- **OT-CFM**: all 7 fixed-crop and 7 refund-jet distributions and both E -shape joints within the JEWEL-vs-JEWEL floor; input correlations 0.91-0.99; no soft floor or flattening above 0.01 GeV (pre-registered reading: useful)
- **α -DSBM** (ε 0.25): toward JEWEL in 6 of 7 and both joints, but 5-23 $\times$ the floor; large per-jet randomness
- below 0.01 GeV both fill 10-15 of JEWEL's 20 empty cone towers with ~ 0.002 GeV (0.1% of E): a limit of $\log(E + 0.1)$



Validation every 10 min (**OT-CFM** at 32 NFE). **α -DSBM**'s refinement worsened its pretrained distributions (mean 0.10 $\rightarrow$ 0.14) while tightening the coupling; E and mass still improving at 2 h.

Not established

- a per-jet medium modification: no truth exists; the map is OT's least change in $\log E$, and its high input correlation is partly by construction
- physics in the energy change: it follows the samples' spectra, in a crop without out-of-cone energy
- robustness: one seed; **α -DSBM**'s limit at a larger budget

One follow-up: **OT-CFM** seeds 1 and 2 (same data, cost, budget, 128 NFE): is the *per-jet* map reproducible, not only its distributions?